Literature Review Guideline

Introduction:

Describe the high level-problem (Research question) of the project:
Breakdown the project into multiple sub-problems for deeper analysis:

General Review

Go through textbooks, wikipedia, web-pages or ask ChatGPT/Gemini to gather information to understand the key concepts and terminologies relevant to the problem and list the concept and relevant papers below.

- Identify Major Ideas:
- Datasets available:
- Common models used for the problem:
- Identify Major issues with the problem/model/datasets
- Also maintain a list of articles/documents for reference for later

Paper Review

Using Appropriate terminologies learnt from above, search for relevant papers using search engines such as : google scholar, semantic scholar. If you hit a paywall, you can find the DOI number for the paper and find the paper in sci-hub.se.

For system design: patents, white papers and product reports can also be useful. Use Google Patents and internet search to look for them. Also Identify if there are any technical standards (procedures) in place for the problem currently in use. For example, teaching standards can be helpful for improving product requirements.

Create a list of papers and track them using Zotero, Mendeley or other Citation Manager. You can use them for annotating and sharing the annotation with the group as well.

There will be a large number of papers available and you'd definitely won't read all of them. Hence, you'd need a strategy to evaluate and select the most relevant, and valuable paper and strategy to read a paper in depth to understand the key ideas. With effective strategy you can go through 5-10 papers in a week (5-10 papers are minimum research to have basic knowledge about the field, not enough to be at the cutting edge.). There are multiple approaches here are some popular suggestions

- Read Papers in a parallel fashion, i.e. try to tackle multiple papers at a time rather than going serially.
- Take multiple pass approach, rather than reading a paper from first word to the last:
    - First Pass to get a **general sense about the paper**: read title, abstract, check figures (architecture, experiment results). In many DL papers 1-2 figure summarize the entire paper. What is the author trying to accomplish
    - Second Pass **to clarify the main idea and contribution of the paper**: Read Intro, Conclusion and deeper dive into figure, Also skim through the related work/literature review in paper to highlight work done by others. Go through the paper and skim the part that don't make sense and try to identify What are the key elements of the approach? What other references can you follow?
    - Third Pass **to get a deeper understanding of the paper to be able to implement them if needed:** Go through the math understanding the overall process and if code is available check it out as well.
    - You can design the number of passes that you want to go through based on your reading patterns.
    - Andrew NG advice on reading research papers and career advice

In the next step for the relevant topics, categorize them into certain categories based on models, methods, datasets, applications, and list them below. You'll be diving deeper into each research paper.

For Literature review report submit a a literature review of at least 5 papers with following contents

- Paper Title , Author, DOI, Link if available publicly
- Main Idea of the paper, or what research question the author is tackling and what contribution they have made
- Dataset(s) used, or strategy for data collection, link (if available)
- Model(s) used, pre-trained model link (if available)
- Code base link (if available)
- Procedure/configuration or test
- Particular techniques/tricks applied
- Performance metrics used and what performance have they achieved
- Your key learning/finding from the paper

Literature Review Matrix

Title/Author/ Date	Conceptual Framework	Research Question(s)/ Hypotheses	Datasets	Methodology	Analysis & Results	Conclusio ns	Implication s for Future Research

Reference: https://academicguides.waldenu.edu/ld.php?content_id=615424

Project Solution Proposals:

Based on the above findings describe how you are planning to conduct your research/experiment and system designing for developing the system.
Update and submit the requirement document if needed with new knowledge available to your from your research.
Design a tentative architecture diagram for the proposed system.

References:

Use Zotero, Mendeley or other Citation Manager to list the sources cited throughout the literature review in a consistent citation style (e.g., APA), this will be helpful later to create a paper or report for your project.